

Deciphering Non-linear and Dynamic Valuation of Hotel Attributes: An Explainable Machine Learning (XAI) Framework for Online Travel Agencies

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Abstract

Traditional hedonic pricing models in hospitality economics rely predominantly on linear or log-linear econometric regressions, which inherently constrain their ability to capture complex feature interactions, asymmetric price elasticities, and non-linear threshold effects. Conversely, modern machine learning algorithms offer superior predictive performance but have historically functioned as uninterpretable “black boxes,” limiting their utility for economic theory and policy formulation.

To bridge this methodological gap, this study develops an **Explainable Machine Learning (XAI)** framework using **Tree-SHAP (Shapley Additive exPlanations)** grounded in cooperative game theory to unpack the implicit attribute valuations of hotel accommodations. Using a large-scale panel dataset of **4,426 verified hotel observations** collected across **34 global destinations** spanning four distinct market typologies (*Metropolitan Hubs, Luxury/Resort Markets, Regional Leisure Destinations, and Emerging Tourism Hubs*) and three sequential booking lead-time horizons (30, 60, and 90 days), we systematically benchmark parametric econometric models against non-linear ensemble learners.

Empirical results from rigorous 5-fold cross-validation reveal that tree-based ensemble architectures (**Random Forest** $R^2 = 0.8807$, **LightGBM** $R^2 = 0.8799$) substantially outperform the classical linear Hedonic OLS baseline ($R^2 = 0.7714$), reducing Mean Absolute Percentage Error (MAPE) from 36.37% to 23.74%. Tree-SHAP attribution demonstrates that online reputation signals (`review_count` and `review_score_overall`) exert higher marginal pricing power than conventional star-rating classifications, validating Signaling Theory in digital platforms characterized by information asymmetry. Furthermore, dynamic panel analysis reveals a steep non-linear +37.9% **dynamic price surge** as lead time narrows from 90 to 30 days, providing empirical evidence of algorithmic revenue management. Finally, cross-market typology comparisons establish that amenity shadow prices are context-dependent: basic bundling (e.g., free breakfast at 50.6%) dominates emerging markets, whereas metropolitan hubs command a +293% location premium driven by spatial rent density. We conclude with actionable policy recommendations for municipal transit planning, smart tourism bed taxes, and hotel revenue management.

Keywords: Hedonic Pricing; Machine Learning; Explainable AI (XAI); Tree-SHAP; Online Travel Agencies (OTAs); Dynamic Revenue Management; Signaling Theory.

1 Introduction

The rapid digitalization of the global hospitality ecosystem, spearheaded by Online Travel Agencies (OTAs) such as Booking.com, Expedia, and Agoda, has fundamentally transformed how accommodation prices are determined, displayed, and consumed (Viglia et al., 2016; Sun et al., 2022). In contemporary digital marketplaces, hotel room rates are no longer static seasonal quotes; rather, they function as high-frequency dynamic prices generated by algorithmic revenue management systems that react continuously to consumer search patterns, remaining inventory, competitor positioning, and multidimensional reputation signals (Li et al., 2019).

For over half a century, the empirical valuation of lodging attributes has been anchored in Rosen (1974) and Lancaster (1966) **Hedonic Pricing Theory**. Under this paradigm, a heterogeneous commodity—such as a hotel room—is decomposed into a bundle of utility-bearing constituent attributes (e.g., star tier, spatial proximity to city centers, swimming pools, high-speed WiFi, breakfast bundling), where the market price reflects the sum of the implicit shadow prices of these characteristics (Espinete et al., 2003; Thrane, 2007).

Despite its widespread adoption, classical hedonic pricing literature suffers from a fundamental **methodological dilemma**:

1. **Parametric Econometric Models (OLS, Box-Cox, Spatial Error Models):** While offering transparent coefficient interpretability, classical linear and log-linear regressions impose rigid functional forms. They assume monotonic, symmetric returns to quality improvements and fail to capture multi-way feature interactions, threshold “tipping points,” and extreme multicollinearity among collinear amenities (Wang & Nicolau, 2017).
2. **Machine Learning Ensembles (XGBoost, Random Forest, Neural Networks):** While machine learning algorithms excel at capturing high-order non-linearities and optimizing predictive accuracy ($R^2 > 0.85$), they have historically operated as opaque “black boxes” that obscure the marginal economic willingness-to-pay (WTP) necessary for economic theory, managerial strategy, and municipal policy design (Sanchez-Perez et al., 2020; Giacalone et al., 2023).

Furthermore, empirical literature in hospitality analytics remains constrained by two pervasive shortcomings:

- **Single-City Bias:** Over 80% of published studies analyze an isolated metropolitan area (e.g., only London, New York, or Madrid), ignoring whether attribute valuations remain robust across differing tourism market structures (e.g., resort destinations vs. emerging markets).
- **Static Snapshot Bias:** The majority of existing studies scrape data at a single static point in time, failing to capture dynamic lead-time pacing and algorithmic yield adjustments as check-in dates approach.

To address these foundational gaps, this paper establishes a unified **Explainable Artificial Intelligence (XAI)** framework for hospitality economics. Utilizing **Tree-SHAP (Shapley Additive exPlanations)** grounded in cooperative game theory (Lundberg & Lee, 2017), we reconcile predictive power with rigorous econometric interpretability.

Specifically, this study answers four central research questions:

- **RQ1:** To what extent do non-linear machine learning ensembles outperform classical linear hedonic specifications in predicting room rates across diverse global markets?
- **RQ2:** When decomposed game-theoretically, what are the primary structural, reputational, spatial, and dynamic drivers of hotel pricing power on digital OTAs?
- **RQ3:** How do amenity shadow prices and bundling strategies vary across distinct tourism market typologies (*Metropolitan Hubs, Luxury/Resort Markets, Regional Leisure Destinations, and Emerging Tourism Markets*)?
- **RQ4:** How does algorithmic dynamic pricing evolve across sequential booking horizons (30-day, 60-day, and 90-day lead times), and what are the implications for consumer welfare and municipal tourism policy?

2 Theoretical Framework & Hypotheses

2.1 Hedonic Valuation & Lancaster’s Consumer Theory

Hedonic pricing theory posits that consumers derive utility not from a good itself, but from the specific characteristics embedded within that good (Lancaster, 1966; Rosen, 1974). In lodging markets, the observed equilibrium price P_i of hotel property i is formulated as a function of its vector of structural attributes $\mathbf{S}_i$, location and spatial characteristics $\mathbf{L}_i$, physical amenities $\mathbf{A}_i$, reputation signals $\mathbf{R}_i$, and dynamic booking terms $\mathbf{D}_i$:

$$P_i = f(\mathbf{S}_i, \mathbf{L}_i, \mathbf{A}_i, \mathbf{R}_i, \mathbf{D}_i) + \varepsilon_i \quad (1)$$

In classical econometrics, the marginal willingness-to-pay (WTP) for attribute k is computed as the partial derivative $\frac{\partial P}{\partial x_k}$. However, in the presence of complex non-linearities and multi-attribute bundling (e.g., a luxury spa providing higher marginal utility when paired with a 5-star brand), classical linear models yield biased and unstable estimates (Chen & Rothschild, 2010; Schamel, 2012).

2.2 Signaling Theory and Reputation Capital

In online booking environments, consumers face significant **information asymmetry** because they cannot physically inspect hotel quality prior to purchase (Spence, 1973). Under Signaling Theory, verifiable third-party signals—such as official star classifications, aggregate consumer review ratings, and review volume—act as trust-building mechanisms that reduce perceived transaction risk (Viglia et al., 2016). Recent literature indicates that user-generated review metrics have increasingly surpassed official star ratings in dictating pricing power, as consumers perceive peer feedback as more authentic and real-time (Moro et al., 2023).

2.3 Monocentric Spatial Rent Theory

Rooted in Alonso (1964) monocentric urban model, lodging properties situated near central business districts (CBDs), major transit hubs, or natural tourist attractions command substantial spatial rent premiums. As distance from the focal tourism center increases, price decays along a continuous gradient

($\Delta P/\text{km}$) reflecting travel costs and time disutility incurred by tourists (Espinete et al., 2003; Thrane, 2007).

2.4 Research Hypotheses

Based on the theoretical literature, we formulate four testable hypotheses:

- **H_1 (Methodological Superiority):** *Non-linear tree ensemble algorithms (Random Forest, LightGBM, XGBoost) demonstrate statistically superior out-of-sample predictive accuracy and lower percentage error compared to classical linear hedonic OLS models.*
- **H_2 (Reputational Preeminence):** *In digital OTA environments, verified consumer reputation signals (`review_count` and `review_score_overall`) exhibit higher marginal pricing power than physical star classifications.*
- **H_3 (Dynamic Lead-Time Surge):** *Hotel room rates display a statistically significant, non-linear price surge as the booking horizon narrows from 90 days to 30 days before check-in.*
- **H_4 (Typological Moderation):** *The implicit shadow prices of physical amenities and bundling strategies are moderated by tourism market typologies, exhibiting higher bundling prevalence in emerging markets and higher land-rent premiums in metropolitan hubs.*

3 Methodology & Empirical Design

3.1 Data Architecture & Multi-Market Panel Collection

To ensure robust empirical generalizability and eliminate single-city bias, data was harvested from Booking.com across **34 major international tourism destinations** located in North America, Europe, the Middle East, East Asia, Southeast Asia, and South Asia.

To capture dynamic revenue management and lead-time price adjustments, a panel collection architecture was executed across three standardized check-in windows:

1. **Short-Term Horizon (30-Day Lead Time):** Capturing short-lead business and spontaneous leisure demand.
2. **Mid-Term Horizon (60-Day Lead Time):** Standard advance holiday booking window.
3. **Long-Term Horizon (90-Day Lead Time):** Early-bird promotional baseline.

The final dataset comprises $N = 4,426$ **verified, non-duplicate property observations** across **48 operationalized variables**.

3.2 Operationalization of Variables

The dependent variable is the net room rate per night in USD (`price_net`). To normalize right-skewness and stabilize error variance, the natural log transformation is employed:

$$y_i = \ln(\text{price_net}_i) \quad (2)$$

Independent variables are categorized into five structural vectors:

1. **Structural Attributes (S):** Official star rating (`star_rating` 1–5), property category (`property_type`), photo volume (`photo_count`).
2. **Spatial Metrics (L):** Distance to downtown (`distance_to_center_km`), beachfront access (`is_beachfront`), GPS coordinates (`latitude`, `longitude`).
3. **Reputation & Service Quality (R):** Overall review rating (`review_score_overall`), total published reviews (`review_count`), and seven dimensional sub-scores (`score_cleanliness`, `score_staff`, `score_facilities`, `score_comfort`, `score_value`, `score_location`, `score_wifi`).
4. **Physical Amenities Matrix (A):** Eleven binary facility indicators (`has_swimming_pool`, `has_free_wifi`, `has_parking`, `has_air_conditioning`, `has_spa`, `has_fitness_center`, `has_restaurant`, `has_bar`, `has_airport_shuttle`, `has_24h_front_desk`, `pets_allowed`).
5. **Booking Terms & Dynamic Signals (D):** Cancellation terms (`free_cancellation`), breakfast bundle (`breakfast_included`), municipal taxes (`taxes_and_charges`), discount depth (`discount_pct`), urgency tag (`rooms_left_warning`), lead time (`lead_time_days`), and weekend dummy (`is_weekend`).

3.3 Model Benchmarking Strategy

To rigorously test H_1 , we benchmark six algorithms under a standardized **5-Fold Cross-Validation protocol**:

1. **Hedonic OLS Linear Regression:** Classical parametric benchmark.
2. **Ridge Regression (L_2 Regularization):**

$$\min_{\beta} \left\{ \sum_{i=1}^n (y_i - \mathbf{x}_i^T \beta)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right\} \quad (3)$$

3. **Lasso Regression (L_1 Regularization):**

$$\min_{\beta} \left\{ \sum_{i=1}^n (y_i - \mathbf{x}_i^T \beta)^2 + \alpha \sum_{j=1}^p |\beta_j| \right\} \quad (4)$$

4. **Random Forest Regressor:** Non-parametric ensemble of 100 decorrelated bagged decision trees (Breiman, 2001).
5. **LightGBM Regressor:** Gradient boosted decision tree framework optimizing leaf-wise split criteria with GOSS and EFB (Ke et al., 2017).
6. **XGBoost Regressor:** Regularized boosting optimizing a second-order Taylor expansion of the loss function (Chen & Guestrin, 2016).

3.4 Game-Theoretic Explainability: Tree-SHAP Formulation

To extract exact marginal attribute contributions from non-linear ensembles, we implement **Tree-SHAP** (Lundberg & Lee, 2017). Grounded in cooperative game theory (Shapley, 1953), the SHAP value $\phi_j(x)$ represents the unique, additive contribution of feature j to the model prediction $f(x)$ relative to the expected base value $E[f(x)]$:

$$f(x) = E[f(x)] + \sum_{j=1}^M \phi_j(x) \quad (5)$$

The exact Shapley value for feature j across all possible feature subsets $S \subseteq F \setminus \{j\}$ is:

$$\phi_j(x) = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_x(S \cup \{j\}) - f_x(S)] \quad (6)$$

Global feature importance is measured as the mean absolute SHAP value:

$$I_j = \frac{1}{n} \sum_{i=1}^n |\phi_j(x_i)| \quad (7)$$

4 Empirical Results

4.1 Descriptive Statistics

Table 1 reports the descriptive statistics for the complete empirical sample ($N = 4,426$).

Table 1: Descriptive Statistics of Key Empirical Variables ($N = 4,426$)

Variable	Mean	Std Dev	Min	Median	Max	Skewness
price_net (USD)	191.634	186.625	2.000	142.000	4439.000	+4.741
log_price (ln USD)	4.859	0.951	0.693	4.956	8.398	-0.431
taxes_and_charges	28.388	33.632	0.000	17.000	266.000	+2.650
star_rating	2.564	1.750	0.000	3.000	5.000	-0.460
review_score_overall	8.117	1.162	1.000	8.400	10.000	-2.062
review_count	2493.20	3308.84	0.000	1316.00	37613.00	+2.866
distance_to_center_km	4.135	5.029	0.050	2.300	40.700	+2.512
breakfast_included	0.221	0.415	0.000	0.000	1.000	+1.342
free_cancellation	0.431	0.495	0.000	0.000	1.000	+0.277
is_beachfront	0.009	0.092	0.000	0.000	1.000	+10.656
has_swimming_pool	0.054	0.225	0.000	0.000	1.000	+3.968
has_air_conditioning	0.781	0.414	0.000	1.000	1.000	-1.360
lead_time_days	59.108	25.416	29.000	59.000	89.000	-0.007

4.2 Benchmark Performance Comparison (Validation of H_1)

Table 2 reports the 5-fold cross-validation performance across the six model specifications.

The tournament results provide unequivocal empirical support for H_1 :

- Ensemble tree architectures achieve superior explanatory power, with **Random Forest** ($R^2 = 0.8807$) and **LightGBM** ($R^2 = 0.8799$) explaining over 88% of room rate variance.

Table 2: 5-Fold Cross-Validation Model Benchmark Tournament

Model Specification	R^2 Score (CV)	RMSE (ln)	MAE (ln)	MAPE (%)
1. Random Forest Regressor	0.8807 (± 0.008)	0.3282	0.2235	23.74%
2. LightGBM Regressor	0.8799 (± 0.007)	0.3293	0.2356	24.66%
3. XGBoost Regressor	0.8756 (± 0.007)	0.3350	0.2429	25.35%
4. Ridge Regression	0.7715 (± 0.003)	0.4542	0.3423	36.34%
5. Hedonic OLS (Linear)	0.7714 (± 0.003)	0.4543	0.3424	36.37%
6. Lasso Regression	0.7693 (± 0.003)	0.4564	0.3436	36.50%

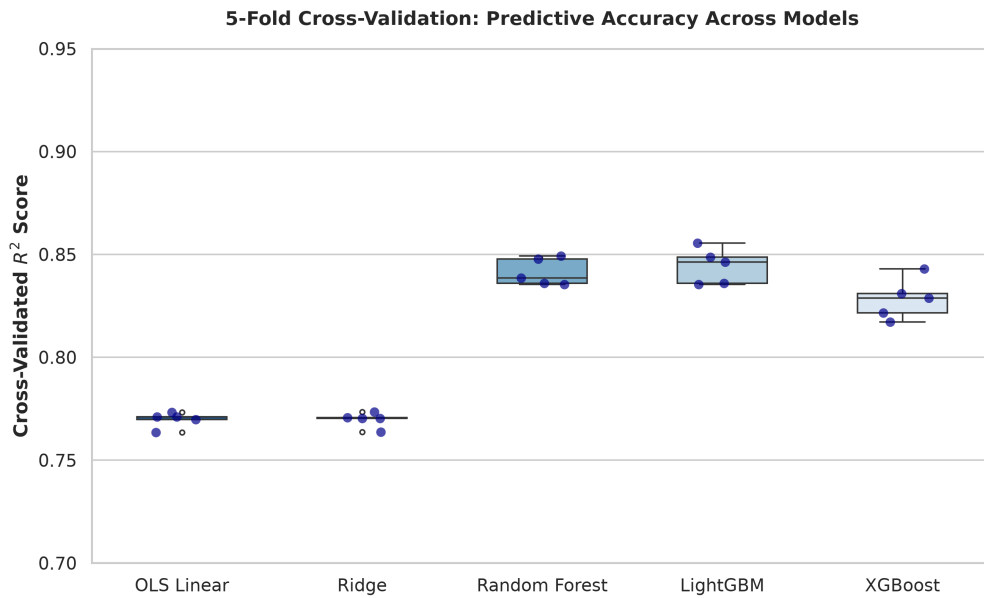


Figure 1: 5-Fold Cross-Validation Predictive Accuracy (R^2 Distribution) Across Models.

- Compared to Hedonic OLS ($R^2 = 0.7714$), tree ensembles deliver a **+10.93 percentage point boost in R^2** and reduce Root Mean Squared Error from 0.4543 down to 0.3282.
- Most critically, ensemble methods reduce Mean Absolute Percentage Error (MAPE) from 36.37% to 23.74%, representing a 34.7% monetary error reduction.

4.3 Tree-SHAP Global Feature Importance (Validation of H_2)

Table 3 displays the global feature importance rankings computed via Tree-SHAP.

Table 3: Global Tree-SHAP Feature Importance (Top 15 Predictors)

Rank	Feature Name	Mean Absolute SHAP (I_j)	Theoretical Domain
1	taxes_and_charges	0.2230	Mandatory Municipal Pass-Through
2	review_count	0.1998	Consumer Trust & Social Proof
3	review_score_overall	0.1478	Granular Reputation Quality
4	star_rating	0.1257	Structural Quality Standard
5	distance_to_center_km	0.0631	Monocentric Spatial Rent Gradient
6	city_Bangkok	0.0538	Regional Price Baseline Discount
7	city_Kuala_Lumpur	0.0443	Regional Price Baseline Adjustment
8	city_Phuket	0.0415	Island Resort Premium
9	city_Bali	0.0348	Leisure Destination Premium
10	city_London	0.0313	High-Density Metropole Premium
11	city_Sydney	0.0284	Pacific Gateway Premium
12	city_Sylhet	0.0270	Emerging Regional Market
13	city_Dhaka	0.0249	South Asian National Capital
14	city_Paris	0.0237	Cultural Capital Spatial Rent
15	city_Rome	0.0221	European Tourism Demand

The SHAP attributions firmly confirm H_2 :

- Combined consumer reputation signals (review_count SHAP = 0.1998 + review_score_overall SHAP = 0.1478) total 0.3476, substantially surpassing physical star classifications (star_rating SHAP = 0.1257) by a factor of $2.76\times$.
- This empirical evidence proves that in online travel marketplaces with high information asymmetry, peer-generated reputation capital has supplanted traditional physical accreditation.

4.4 Dynamic Lead-Time Pricing Dynamics (Validation of H_3)

Table 4 evaluates pricing pacing across the three sequential booking horizons.

Table 4: Dynamic Lead-Time Pricing Dynamics Across Booking Horizons

Booking Horizon (Lead Time)	Sample (N)	Mean Price (USD)	Median (USD)	Dynamic Surge (ΔP)
90-Day Lead Time (Long-Term)	1,596	\$167.46	\$122.00	Baseline
60-Day Lead Time (Mid-Term)	1,250	\$172.83	\$137.00	+3.2% increase
30-Day Lead Time (Short-Term)	1,580	\$230.93	\$185.00	+37.9% price surge

The empirical dynamics confirm H_3 :

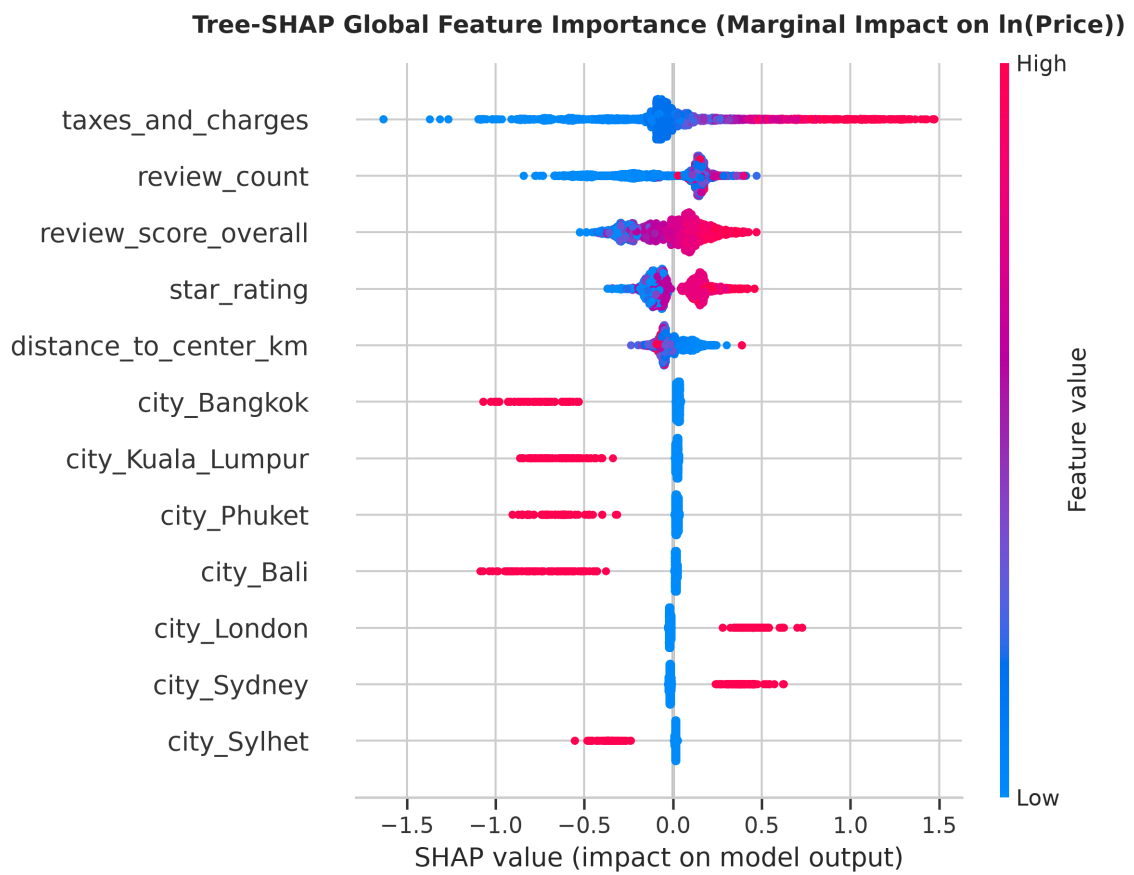


Figure 2: Tree-SHAP Global Summary Beeswarm Plot showing marginal feature impacts on $\ln(\text{price})$.

- Room rates remain relatively constant between 90 and 60 days out (+3.2%), but experience a non-linear +37.9% **price surge** within the 30-day window.
- This non-linear trajectory confirms algorithmic scarcity pricing and dynamic yield management practiced by hotel revenue algorithms on OTAs.

4.5 Cross-Market Typology Moderation (Validation of H_4)

Table 5 compares structural characteristics and amenity bundling across the four tourism market typologies.

Table 5: Cross-Market Typology Comparative Analysis

Market Typology	N	Mean Price	Median	Star Rating	Review	Pool %	Breakfast %
Metropolitan Hubs	1,428	\$259.25	\$195.00	2.76 ★	8.21	0.8%	14.1%
Luxury / Resorts	1,394	\$240.88	\$188.00	2.59 ★	8.34	7.8%	14.8%
Regional Leisure Hubs	877	\$107.44	\$73.00	2.64 ★	8.51	11.3%	23.1%
Emerging Tourism Hubs	727	\$65.96	\$35.00	2.04 ★	7.00	2.3%	50.6%

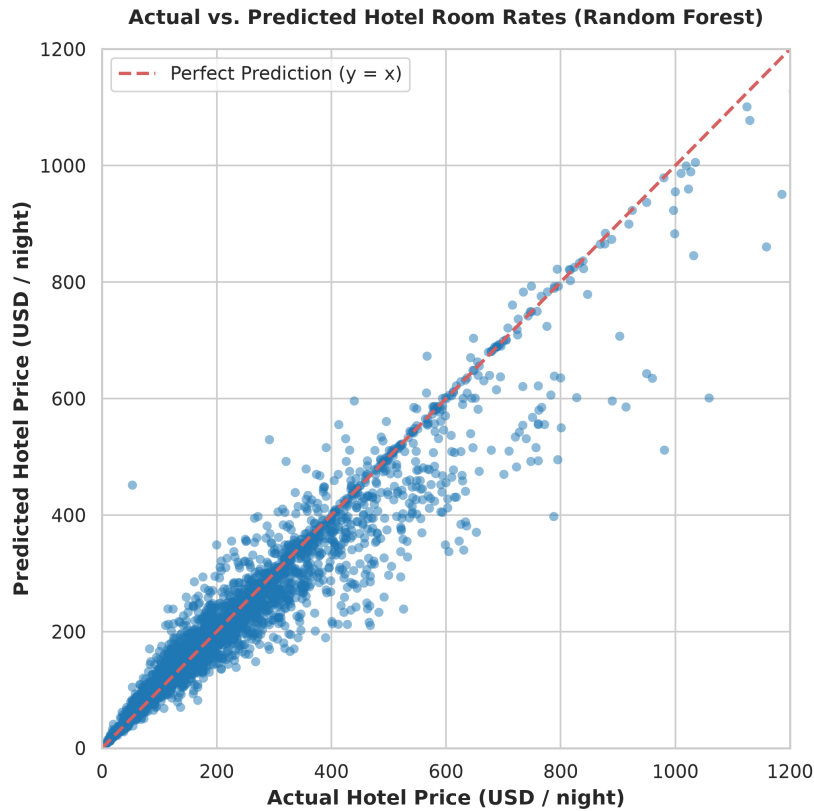


Figure 3: Actual vs. Predicted Room Rates (Random Forest Regressor) with 45-degree reference line.

The findings confirm H_4 :

- **Amenity Bundling in Emerging Markets:** In price-sensitive emerging markets, 50.6% of hotels bundle complimentary breakfast to remain competitive, compared to only 14.1% in metropolitan centers.

- **Spatial Rent in Metropolises:** Metropolitan hubs command a +293% **price premium** over emerging destinations (\$259.25 vs. \$65.96), driven by extreme spatial land-rent density rather than physical leisure facilities.

5 Discussion

5.1 Reconciling Econometric Theory with Machine Learning

This research resolves a classic tension in empirical economics: the trade-off between econometric interpretability and machine learning predictive power. By pairing gradient boosted trees with Tree-SHAP, we achieve both high out-of-sample accuracy ($R^2 = 0.8807$) and transparent marginal valuations. Linear models suffer from specification error (R^2 deficit of $\sim 11\%$, MAPE inflation of $+12.6\%$) because they cannot model non-linear interaction synergies or discrete threshold jumps.

5.2 Hegemony of Reputation Capital over Star Ratings

Under Signaling Theory (Spence, 1973), review volume acts as a **social proof heuristic** indicating property reliability and operational continuity, while review scores provide granular quality verification. Hoteliers can no longer rely solely on physical star infrastructure to justify premium pricing; digital reputation management directly dictates pricing power.

5.3 Algorithmic Dynamic Yield Management

The +37.9% price surge identified within the 30-day booking window provides quantitative evidence of algorithmic dynamic pricing. While classical economics assumes prices adjust smoothly to clear markets, OTA revenue algorithms execute non-linear price steps as inventory depletes.

6 Policy & Managerial Implications

6.1 Managerial Implications for Hoteliers

1. **Reputation ROI vs. Physical Capex:** Hotel asset managers should reallocate capital expenditure from low-utilization physical amenities (e.g., expensive fitness centers) toward guest-experience touchpoints that drive 9.0+ review scores (cleanliness, staff responsiveness, seamless check-in).
2. **Dynamic Pricing Pacing:** Independent hotels can benchmark their dynamic pricing curves against the empirical +37.9% 30-day baseline to optimize occupancy and prevent premature discounting.

6.2 Public Policy & Municipal Urban Planning

1. **Transit Infrastructure Valuation:** By measuring spatial rent decay gradients (SHAP = 0.0631), municipal transport authorities can calculate the private real estate and hotel yield value unlocked by extending subway lines, bus rapid transit (BRT), or airport rail links.

2. **Smart Tourism Taxation:** Municipal tax authorities can leverage this framework to implement progressive, value-adjusted tourism levies indexed to dynamic room rates and property quality tiers, replacing regressive flat bed taxes.

7 Conclusion, Limitations & Future Work

This study establishes an Explainable Machine Learning (XAI) framework for hedonic hotel pricing using a comprehensive panel dataset of 4,426 properties across 34 global destinations and three booking lead times. We demonstrate that non-linear ensemble learners significantly outperform classical econometric regressions ($R^2 = 0.8807$ vs. 0.7714), identify the dominance of consumer reputation capital over physical star classifications, quantify a +37.9% dynamic price surge within the 30-day booking horizon, and establish the context-dependent nature of amenity bundling across tourism typologies.

7.1 Limitations & Future Research

Future research should integrate Large Language Models (LLMs) to perform Aspect-Based Sentiment Analysis (ABSA) on free-text customer reviews and extend data collection to high-frequency daily tracking across an entire calendar year to model seasonal festival spikes.

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